

SAHIL KHOKHAR

khokhar.sahil0809@gmail.com | Toronto, ON | salkhokhar.github.io | linkedin.com/in/sahil-khokhar-5b1610165 | github.com/salkhokhar

SUMMARY

AI engineer who builds production LLM systems: agentic orchestration with tool calling and human-in-the-loop control, MCP tool platforms with role-based access, and privacy-preserving infrastructure for generative AI. PhD (computer vision) with five peer-reviewed publications on YOLO-based object detection and CNN-based character recognition.

SKILLS

AI / LLMs: Azure OpenAI, OpenAI API, agentic AI, LLM orchestration, tool calling, human-in-the-loop, MCP, NLP, Transformers

ML & Computer Vision: PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, OCR (EasyOCR)

Programming: Python, SQL, C#, .NET, C

Backend, DevOps & Data: FastAPI, REST APIs, Docker, Git, CI/CD, PostgreSQL, SQL Server, Redis

Privacy & Security: Presidio, spaCy, PII masking and re-identification, RBAC

EXPERIENCE

CGI, Toronto, ON

Sep 2025 – Present

AI Developer (Automation, Data & AI Initiatives), Insurance Information Services

- Built an MCP server platform that exposes analytics and data-visualization tools to clients through role-based access control.
- Designed an AI privacy gateway for secure LLM use: PII masking and re-identification with Presidio, spaCy, and regex, a secure token vault, and OCR-based ingestion of PDFs and images using EasyOCR.
- Built FastAPI services on Azure OpenAI that convert unstructured requests into structured, schema-validated tickets, with duplicate detection and CI/CD integration.
- Built an agentic workflow on Azure OpenAI that classifies inbound service requests, triggers automated actions, and requests human approval before acting, with full run logging and auditability.

First National Financial LP, Toronto, ON

May 2025 – Aug 2025

Software Engineer (Co-op)

- Built an AI-powered underwriting assistant using OpenAI APIs to support decision-making workflows.
- Integrated multiple enterprise data sources into unified AI pipelines and implemented LLM-based extraction with Azure OpenAI to transform unstructured data into structured, validated schemas.
- Improved underwriting efficiency through automated data retrieval, validation, and fault-tolerant processing; packaged services with Docker.

Guru Jambheshwar University of Science and Technology, India

Nov 2018 – Aug 2024

Research Scholar (PhD), Electronics and Communication Engineering

- Conducted applied AI and computer vision research focused on object detection and license plate recognition systems.
- Developed and optimized machine learning models using TensorFlow and PyTorch.
- Implemented and evaluated multiple YOLO-based architectures for real-time object detection in images and video streams.

EDUCATION

PhD, Electronics & Communication Engineering, Guru Jambheshwar University of Science and Technology, India

Thesis: Advanced Character Detection and Recognition Techniques using Soft Computing Algorithms

M.Eng, Electrical and Computer Engineering (Applied AI concentration), University of Ottawa, Canada

M.Tech, Electronics & Communication Engineering, Deenbandhu Chhotu Ram University of Science and Technology, India

Thesis: SVM-Based Character Recognition for ALPR Systems

B.Tech, Electronics & Communication Engineering (First Division), Maharshi Dayanand University, India

PUBLICATIONS

1. **S. Khokhar**, D. Kedia. "Integrating YOLOv8 and CSPBottleneck based CNN for enhanced license plate character recognition." *Journal of Real-Time Image Processing*, 21(5):168, 2024. DOI
2. **S. Khokhar**, D. Kedia. "Improving license plate detection with YOLO-LPD algorithm." *International Journal of Computational Vision and Robotics*, 2024. DOI
3. **S. Khokhar**, D. Kedia. "Enhanced licence plate detection using YOLO framework in challenging environments." *International Journal of Computational Vision and Robotics*, 2024. DOI
4. **S. Khokhar**, D. Kedia, P. K. Dahiya. "License plate detection techniques: conventional methods to deep learning." *ICT with Intelligent Applications (Smart Innovation)*, 2022. DOI
5. **S. Khokhar**, P. K. Dahiya. "Character recognition for ALPR systems: a new perspective." *Innovations in Electronics and Communication Engineering*, pp. 479-485, 2020. DOI